PSTAT 5A | Quiz 4 Solutions

Given: The weight of an adult male Channel Island fox is normally distributed with mean 4.8 pounds and variance 0.25 pounds squared.

(a) Define the random variable of interest, X.

Let X be the weight, in pounds, of one randomly selected adult male Channel Island fox.

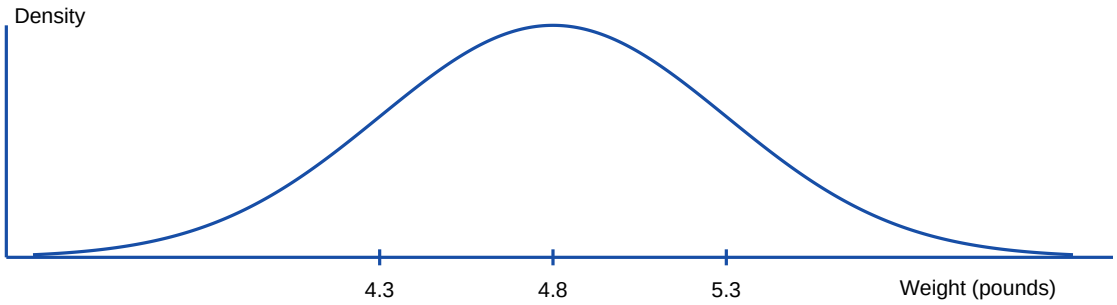
(b) State the distribution of X.

The mean is 4.8 and the variance is 0.25. Therefore, the standard deviation is the square root of 0.25, which is 0.5 pounds.

$X \sim N(\mu = 4.8, \sigma = 0.5)$

(c) Sketch the density curve, with appropriate labels.

Draw a symmetric bell-shaped normal density curve centered at x = 4.8. Label the horizontal axis "Weight (pounds)" and the vertical axis "Density". Mark 4.8 at the peak, and optionally mark 4.3 and 5.3 (one standard deviation below and above the mean). The total area under the curve is 1.



(d) What is the probability that its weight is at most 4.3 pounds?

"At most" means X is less than or equal to 4.3. Standardize using Z = (X - mean) / standard deviation.

P(X <= 4.3) = P(Z <= (4.3 - 4.8) / 0.5) = P(Z <= -1.0).

By symmetry of the standard normal distribution, P(Z <= -1.0) = 1 - P(Z <= 1.0).

Using the normal table: 1 - 0.8413 = **0.1587** (about 15.87%).

(e) What is the probability that its weight is at least 4 pounds?

"At least" means X is greater than or equal to 4. Standardize:

P(X >= 4.0) = P(Z >= (4.0 - 4.8) / 0.5) = P(Z >= -1.6).

By symmetry, P(Z >= -1.6) = P(Z <= 1.6).

Using the normal table: **0.9452** (about 94.52%).

Note: For a continuous random variable, strict and non-strict inequalities give the same probability at a single cutoff because P(X = a) = 0.